MIL-P-46161A(MR)
25 August 1976
SUPERSEDING
MIL-P-46161(MR)
2 April 1973

MILITARY SPECIFICATION

PLASTIC MOLDING MATERIAL, POLYTEREPHTHALATE THERMOPLASTIC, GLASS FIBER REINFORCED

This specification is approved for use by the Army Materials and Mechanics Research Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers glass fiber reinforced polyterephthalate thermoplastic molding material.

1.2 Classification. The material shall be of the following grades and classes as specified (see 6.2).

Grade A - General purpose

Class 10 - 10 percent nominal glass fiber
Class 15 - 15 percent nominal glass fiber
Class 20 - 20 percent nominal glass fiber
Class 30 - 30 percent nominal glass fiber
Class 40 - 40 percent nominal glass fiber

Grade B - Flame retardant

Class 15 - 15 percent nominal glass fiber
Class 20 - 20 percent nominal glass fiber
Class 30 - 30 percent nominal glass fiber

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 9330

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SPECIFICATIONS

FEDERAL

PPP-D-723 - Drums, Fiber
PPP-D-729 - Drums: Metal, 55-Gallon (for Shipment of Noncorrosive Material)

MILITARY

MIL-P-46160(MR) - Plastic Molding Material, Polyterephthalate, Thermoplastic

STANDARDS

FEDERAL

Fed. Std. No. 356 - Commercial Packaging of Supplies and Equipment

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards

D 149 - Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies
D 150 - AC Loss Characteristics and Dielectric Constant (Permittivity) of Solid Electrical Insulating Materials
D 256 - Impact Resistance of Plastics and Electrical Insulating Materials
D 618 - Conditioning Plastics and Electrical Insulating Materials for Testing
D 638 - Tensile Properties of Plastics
D 648 - Deflection Temperature of Plastics Under Load
D 790 - Flexural Properties of Plastics
D 792 - Specific Gravity and Density of Plastics by Displacement.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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Underwriters' Laboratories (UL), Inc.

UL 94 - Tests for Flammability of Plastic Materials for Parts in Devices and Appliances, dated December 1973

(Application for copies should be made to the Underwriters' Laboratories, Inc., 207 E. Ohio Street, Chicago, IL 60611.)

3. REQUIREMENTS

3.1 Material. The material shall consist of poly(tetramethylene) terephthalate thermoplastic reinforced with glass fiber. The poly(tetramethylene) terephthalate thermoplastic shall conform to type II material specified in MIL-P-46160(MR). Grade B material shall have flame retardants added to the glass reinforced poly(tetramethylene) terephthalate.

3.2 Property values. Specimens prepared from the material, as specified in 4.3.1, shall conform to the property values specified in table I, when tested as specified in the applicable procedure of 4.3. When specified by the procuring activity (see 6.2), the electrical properties specified in table II shall be included as requirements.

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Table I. Property values

Property	Grade A					Grade B		
	Class 10	Class 15	Class 20	Class 30	Class 40	Class 15	Class 20	Class 30
Specific gravity range	1.35-1.40	1.39-1.45	1.41-1.51	1.48-1.59	1.58-1.68	1.51-1.57	1.52-1.65	1.62-1.73
Tensile strength, pounds per square inch (psi), minimum (min.)	10,000	11,500	13,000	15,000	18,500	11,500	13,000	15,000
Flexural modulus, 10^5 psi, min.	4.5	6.5	7.5	10.	15.	7.5	9.	11.5
Izod impact strength[1/], foot, pounds per inch of notch, min. at -40°C (-40°F)	0.4	0.8	0.8	1.0	1.3	0.7	0.7	0.9
Deflection temperature under load at 264 psi fiber stress, min., °C	130	182	180	200	210	182	188	195
°F	266	360	356	390	410	360	370	383
Flammability, 1/32 inch thick specimen	NA[2/]	NA[2/]	NA[2/]	NA[2/]	NA[2/]	94V-0[3/]	94V-0[3/]	94V-0[3/]

1/ Izod impact is included for aid in material identification only and does not predict nor correlate with performance of parts molded of these materials.

2/ NA - not applicable.

3/ 94V-0 by Underwriters' Laboratories test specified in 4.3.8. Correlation with flammability under actual use conditions is not necessarily implied.

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Table II. Electrical property values 1/

Property	Grade A					Grade B		
	Class 10	Class 15	Class 20	Class 30	Class 40	Class 15	Class 20	Class 30
Dielectric constant at 100 Hz, max.	3.7	3.9	3.8	4.3	4.5	3.9	4.0	4.0
Dielectric strength in air or oil, volts per mil, Min. 1/8 inch thick specimen	400	375	425	400	425	370	400	400
1/16 inch thick specimen	550	500	575	575	575	475	525	525

1/ Applicable only when specified by the procuring activity (see 6.2).

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3.3 Color. The color of the molding or extrusion material, including tolerances, and the method of determining compliance shall be as specified by the procuring activity (see 6.2). If color of molded plaques is required, the color and method of determination shall be as specified by the procuring activity (see 6.2). When color is not specified, classes 10, 15, 20, 30 and 40 material shall be off white translucent.

3.4 Form. When required by the procuring activity (see 6.2) the form of the material (particle size and shape) and the method of determining compliance, shall be as specified by the procuring activity. Otherwise, the form normally supplied by the manufacturer shall be acceptable.

3.5 Workmanship. The material shall be clean and contain no foreign material. The material shall be uniform in color, form, and all other appearance as determined visually.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, ... t where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same class, submitted for delivery at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1 ... the supplier is responsible for insuring that materials and components ... e manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent ... ried, or, if none, in accordance with this specification. In the event ... conflict, this specification will govern. A supplier's certificate of compliance with the requirements specified in 3.1 and the actual glass content for each class supplied shall be furnished.

4.2.2 Inspection of material

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4.2.2.1 Examination of the material. Examination of the material shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQL's) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of 100 pounds for examination in 4.2.2.1.1, and in units of shipping containers for examination in 4.2.2.1.2.

4.2.2.1.1 Examination of the material for defects in appearance and workmanship. The sample unit for this examination, specified in table III shall be approximately one pound.

Table III. Examination for defects in color, form, uniformity and appearance

Examine	Defect
Appearance and workmanship	When color specified, color1/ incorrect When color specified, not uniform Material not white opaque, when color not specified Form1/ not as specified Form not uniform Not uniform in other appearance Not clean, presence of foreign material

1/ When specified by the procuring activity (see 3.3 and 3.4).

4.2.2.1.2 Examination of packaging. An examination shall be made in accordance with table IV, to determine that packing and marking comply with section 5 requirements. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examinaed for closure defects.

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Table IV. Examination of packaging

Examine	Defects
Packing	Not level specified; not in accordance with contract requirements. Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability. Inadequate application of components such as: incomplete closures of case liners; container flaps, loose or inadequate strappings, improper taping or inadequate stapling. Bulged or distorted containers.
Quantity of material	Less than specified or indicated quantity.
Weight	Gross weight exceeds specified requirements.
Markings	Interior or exterior markings omitted, illegible, incorrect, incomplete, of improper size, location or sequence or method of application. Not in accordance with contract requirements, additional marking is specified

2.2.1.3 Inspection levels and acceptable quality levels (AQLs) for ...inations. The inspection levels for determining the sample size and ...cceptable quality level (AQL) expressed as defects per 100 units ... be as follows:

...ination paragraph	Inspection level	AQL
4.2.2.1.1	II	2.5
4.2.2.1.2	S-2	2.5

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4.2.3 Testing. The material shall be tested for the applicable characteristics listed in table I and table II (when specified). Testing shall be in accordance with the test methods specified, for each lot submitted for inspection. The lot size for the purpose of determining sample size for testing (see MIL-STD-105) shall be expressed in units of 50 pounds of material. The sample unit of product shall consist of sufficient material to prepare all required specimens. The inspection level shall be S-1, with an acceptance number of 0. The results for each test shall be the averaged results of the specimens, unless only one specimen is required for testing.

4.2.3.1 Classification of tests. All tests shall be classified as follows.

a. Lot acceptance tests (see 4.2.3.2).
b. Periodic lot check tests (see 4.2.3.3).
c. Electrical property tests (see 4.2.3.4).

4.2.3.2 Lot acceptance tests. Lot acceptance tests shall be made on each lot of material and shall be the basis for acceptance or rejection of the lot, except when periodic lot check tests are required. Lot acceptance tests shall consist of testing for specific gravity for grade A material. Lot acceptance tests for grade B material shall consist of testing for specific gravity and flammability.

4.2.3.3 Periodic lot check tests. Periodic lot check tests shall be made on the first lot of material furnished under this specification and on any subsequent lot specified by the procuring agency (see 6.2). If not specified by the procuring agency, periodic lot check tests shall be repeated at least once every year. Periodic lot check tests shall consist of all tests specified in table I. When periodic lot check tests are made, they shall be included in the basis for acceptance or rejection of the lot.

4.2.3.4 Electrical property tests. Electrical property tests shall be made on any lot specified by the procuring agency (see 6.2), and shall be included in the basis for acceptance or rejection of the lot.

4.3 Test methods.

4.3.1 Specimen preparation. Test specimens shall be prepared by injection molding in accordance with recommendations of the manufacturer.

4.3.2 Specimen conditioning and testing. Unless otherwise specified, test specimens shall be conditioned in accordance with procedure A of ASTM D 618 and tested at 23° ± 1°C. (73.4° ± 1.8°F.) and 50 ± 5 percent relative humidity

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4.3.3 Specific gravity. One determination shall be made in accordance with ASTM D 792. Method A1 or A2 of ASTM D 792 shall be used, with the specimen conforming to dimensional requirements of ASTM D 792.

4.3.4 Tensile strength. Three specimens shall be tested in accordance with ASTM D 638, using testing speed B. Dimensions of each specimen shall conform to type I of figure 1 of ASTM D 638.

4.3.5 Flexural modulus of elasticity. Three specimens shall be tested in accordance with ASTM D 790, method 1 and procedure B. The specimen shall be 1/8 inch by 1/2 inch and the central portion of a tensile bar may be used.

4.3.6 Impact strength. Three specimens shall be tested in accordance with ASTM D 256, using method A. Specimens shall be 0.125 by 0.5 inch bar. Testing shall be at -40°C. (40°F.).

4.3.7 Deflection temperature. Three specimens shall be tested in accordance with ASTM D 648. Specimen thickness shall be 0.125 inch.

4.3.8 Flammability. Specimens shall be tested in accordance with Underwriters' Laboratories UL 94 using 1/16 inch thick specimens. The number of specimens to be tested and dimensions for length and width shall be as specified in Underwriters' Laboratories UL 94.

4.3.9 Dielectric constant. The dielectric constant shall be determined on 2 specimens, each 4 inches in diameter and 1/8 inch thick, in accordance with ASTM D 150 using a frequency of 100 Hz.

4.3.10 Dielectric strength. The dielectric strength shall be determined by the "short-time test" of ASTM D 149. Testing shall be either in air or oil using 1 inch electrodes. Either three 1/8 inch thick specimens or three 1/16 inch thick specimens shall be tested.

5. PACKAGING

.1 Packing. Packing shall be level A, B, or Commercial as specified 6.2).

5.1.1 Level A. Unless otherwise specified, the material shall be packed in one of the following types of containers:

a. Fiber drums conforming to PPP-D-723, type II, grade A, or type III, grade A in quantities of 200 pounds, maximum.

b. Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds, maximum.

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Insofar as practical, drums shall be of uniform shape and size, with minimum cube and tare consistent with the protection required. Drums shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with a 0.004 inch thick polyethylene liner properly heat sealed.

5.1.2 Level B. Unless otherwise specified, the material shall be packed in one of the following types of containers:

a. Fiber drums conforming to PPP-D-723, type I, grade A in quantities of 200 pounds maximum.

b. Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds, maximum.

Insofar as practical, drums shall be of uniform shape and size with minimum cube and tare consistent with the protection required. Drums shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with a 0.004 inch thick polyethylene liner properly heat sealed.

5.1.3 Commercial. The material shall be packed in accordance with Fed. Std. No. 356.

5.1.4 Disposability (see 6.3).

5.2 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129, with the exception that commercial marking in accordance with Fed. Std. No. 356 applies for commercial packaging only.

6. NOTES

6.1 Intended use. Glass fiber reinforced poly(tetramethylene) terephthalate injection molding materials are intended for application where a relatively inexpensive engineering type plastic material is required. Applications include housing for light fixtures and electrical drills, terminal notch boxes, photographic drain trays, X-ray machine knobs, gasoline cap vents, and gear protection equipment. The flame retardant grade includes applications such as insulation material for various electrical and electronic connectors. The glass fiber reinforced material is particularly superior to unreinforced material in deflection temperature under load and tensile strength.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Grade and class of material required (see 1.2).
c. Electrical property tests, if required (see 3.2, table II, and 4.2.3.4).
d. Color, if required, including use of molded plaques, if required (see 3.3).
e. Form, if required (see 3.4).
f. Periodic lot check tests, if required (see 4.2.3.3).
g. Level of packing required (see 5.1).

6.3 Disposability of packing materials. Environmental pollution preventative measures are contained in the packing material specifications indicated in section 2. Refer to these specifications for recommended disposability methods.

6.4 Cross reference. Table V shows the classification of material covered by this specification and the corresponding classification of material specified in MIL-P-46161(MR).

Table V. Classification in this document and corresponding classification in MIL-P-46161(MR)

MIL-P-46161A(MR)	MIL-P-46161(MR)
Grade A	Grade A
Class 10	Class 1
Class 15	None
Class 20	Class 2
Class 30	Class 3
Class 40	Class 4
Grade B	Grade B
Class 15	None
Class 20	Class 2
Class 30	Class 3

Custodian:
Army - MR

Review activities:
Army - EL, GL, ME

User activity:
Army - AT

Preparing activity:
Army - MR

Project No. 9330-A699

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR
b. ADDRESS (Street, City, State, ZIP Code)	☐ USER
	☐ MANUFACTURER
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
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